

University Case Competition 2026

Aegis & Aegis

Wound Closure Expansion Strategy

Medical Devices & Diagnostics

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Johnson & Johnson

Team 5 | Rutgers University | April 2026

Our Strategy:

- ❖ **R&D:** Internally Develop Spiral; Purchase IP of Symmetrical
- ❖ **Pricing:** Medium
- ❖ **Marketing:** Silver + Gold
- ❖ **Supply Chain:** Plant Extension
- ❖ **Human Capital:** Hire full time employees

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Executive Summary

Our recommended strategy for A&A's barbed suture expansion

OUR DECISION: Our decision: Option 4 — Develop spiral internally → Purchase IP for symmetrical design, Speed, control, risk, High-growth, high-return, & credo-aligned strategy

FINANCIAL VALUE	MARKET POSITION	CREDO ALIGNMENT
NPV: \$13.21B (7% discount rate) IRR: 222% (32x hurdle rate) ROI: 553.5% Payback: ~3 yrs Break-Even: Yr 4	Peak market share: 84.2% (barbed suture market) First to market in spiral — highest market share potential Defend non-barbed leadership with symmetrical IP purchase	Medium pricing — affordable for 80% of target market Full-time employees — security, dignity & long-term commitment Plant expansion in Cornelia, GA — community & quality investment

Market Overview & SWOT

Wound closure market landscape — barbed suture opportunity

STRENGTHS

- Aegis and Aegis #1: in non-barbed sutures; strong base in \$4.56B global market
- Trusted across 6,100 U.S. hospitals (27M+ surgeries/year); strong GPO/IDN ties
- Existing Cornelia, GA manufacturing lowers CapEx and speeds product launch (time to market)
- \$27.4B Med Devices segment; top 1–2 positions in 80% of platforms
- A purpose-led brand that facilitates purchasing and formulary inclusion

OPPORTUNITIES

- SSI risk (0.5–3%) drives demand for advanced closure solutions
- Barbed sutures reduce closure time (~110 sec) with no added risk
- Proven OR time + cost savings in procedures like spine surgery
- Growth in robotic surgery increases demand for optimized sutures
- First-mover potential in spiral sutures, especially in Asia-Pacific

WEAKNESSES

- No barbed suture offering (missing fastest-growing segment)
- **Cornelia plant near capacity** → requires CapEx or partnerships
- Late entrant vs. competitors with 10+ years clinical adoption
- FDA pathway (510k/PMA) adds 12–18+ months + trial costs
- No spiral/barbed IP; competitor patents limit design flexibility

THREATS

- Competitors (Athena) with strong IP + global presence
- Guidelines favor antimicrobial-coated sutures over barbed sutures
- Alternatives (staples, adhesives) gaining traction
- Global regulatory timelines (18–36 months) delay expansion
- Price pressure from low-cost Asian manufacturers

Competitive Analysis

How A&A stacks up against Crius and Athena in wound closure



Competitive Analysis

How A&A stacks up against Crius and Athena in wound closure

Criteria	A&A (Us)	Crius / Poseidon	Athena / Kronos
Revenue (2012)	\$27.4B MedTech	~\$11.9B	\$221.3M
Market Position	Leader — non-barbed	FDA approved 2010	Patented since 1999
Suture Type	Non-barbed (current)	Dual angle cut	Symmetrical (Kronos)
Geography	Global	55% US / 45% Intl	62% US / 38% Intl
Key Strength	Scale & trust	Large established base	Differentiated design
Key Risk	Late to barbed market	Direct competitor	Growing fast

R&D Strategy Decision

How A&A will enter the barbed suture market

OUR DECISION: Our decision: Option 4 — Develop spiral internally · Purchase IP of symmetrical

Option 1

Purchase IP Both Sutures

✓ PROS

- ✓ Faster to market
- ✓ Guaranteed market share
- ✓ Lower R&D cost

✗ CONS

- ✗ Higher CapEx
- ✗ Not first to market on either
- ✗ Delayed spiral timeline

Option 2

Develop Both Internally

✓ PROS

- ✓ Own all IP
- ✓ First mover on spiral
- Higher market share potential

✗ CONS

- ✗ More R&D spend
- ✗ Late on symmetrical -> lower mkt share
- ✗ Higher risk

Option 3

Buy Spiral IP Develop Sym.

✓ PROS

- ✓ First mover spiral
- ✓ Higher spiral market share
- ✓ Symmetrical guaranteed

✗ CONS

- ✗ Split resources
- ✗ Symmetrical later launch -> lower mkt share
- ✗ R&D intensive
- ✗ Split resources

Option 4

Develop Spiral Buy Sym. IP

✓ PROS

- ✓ First mover spiral
- ✓ Sym. faster launch
- ✓ Own spiral IP
- ✓ Balanced risk

✗ CONS

- ✗ Higher R&D cost
- ✗ Spiral timeline risk
- ✗ Dual investment

Pricing Strategy Decision

Setting the right price to balance access, profit, and competitive position

OUR DECISION: [Low / Medium / High] Pricing

Credo Consideration: 'We must constantly strive to provide value, reduce our costs and maintain reasonable prices.'

LOW

Market Share Growth	100% of target market
Positioning	Below market average
Margin Impact	Minimum viable
Key Tradeoff	Lowest profitability

MEDIUM

Market Share Growth	80% of target market
Positioning	Market average
Margin Impact	Sustainable + growth
Key Tradeoff	Balanced approach

HIGH

Market Share Growth	65% of target market
Positioning	Above competitors
Margin Impact	Maximum profit
Key Tradeoff	Limits accessibility

Marketing Strategy Decision

Reaching hospitals and surgeons with the right level of engagement

OUR DECISION: Silver Marketing Strategy with Targeted Gold Elements

Why Silver + Gold?

- Balanced investment: more impactful than Bronze, more efficient than Gold
- Supports market transition — barbed sutures still in adoption phase
- Aligned with medium pricing — avoids overspending relative to margins

"Silver builds credibility through clinical education, while selective Gold elements accelerate adoption through hands-on surgeon engagement."

Marketing Strategy: Bronze vs. Gold vs. Silver

Why our Silver + targeted Gold approach is the right fit for A&A's barbed suture launch

OUR DECISION: Silver — Balanced Investment Aligned with Medium Pricing & Existing Brand Trust

Why Not Bronze?

- Relies solely on digital outreach — insufficient for a clinical, trust-based market
- Lacks educational depth for a new surgical technique like barbed sutures
- Cannot support the surgeon training required for knotless technique adoption

Why Not Full Gold?

- Highest cost and risk — unjustified given medium pricing model
- Since A&A is licensing IP (not in-house), aggressive spend is unwarranted
- A&A's existing hospital relationships eliminate the need to build from scratch

Target Audience

Reaching hospitals and surgeons with the right level of engagement

Target Customers

- Hospitals & surgical centers
- Surgeons performing minimally invasive and robotic procedures
- Procurement teams and GPOs (Group Purchasing Organizations)

Customer Needs

- Efficiency in surgery — reduced operating time via knotless technique
- Reduced complications — lower infection risk vs. traditional sutures
- Cost-effective solutions — clear cost-per-procedure value
- Clinical confidence — evidence and training before technique switch

How Our Strategy Reaches Them

- Silver → clinical brochures, comparison charts, packets to procurement teams
- Silver → email announcements and product videos to A&A's hospital customer base
- Gold elements → surgical conferences, live demos, trial kits for high-volume centers
- Medium pricing → affordability and accessibility across market segments

Key Adoption Barriers We Address

- Higher cost vs. traditional sutures — mitigated by medium pricing & GPO targeting
- Surgeon training required — addressed via hands-on demos and knotless technique kits
- Low awareness of clinical benefits — tackled through Silver educational materials
- Decision-cycle length — shortened by leveraging A&A's existing hospital relationships

Supply Chain Strategy Decision

How A&A will manufacture its new barbed sutures

OUR DECISION: Plant Extension — Cornelia, GA

Option 1 Outsource

- ✓ Lowest CapEx
- ✓ Immediate revenue (Year 1)
- ✗ No IP / quality control
- ✗ No Human Capital decision
- ✗ Third-party dependency

Option 2 Plant Extension ✓ CHOSEN

- ✓ Moderate CapEx
- ✓ Leverage existing Cornelia plant
- ✓ Full quality & IP control
- ✓ Operational in ~3 years (Year 4)
- ✓ Supports Credo – community & jobs

Option 3 New Plant

- ✓ Maximum capacity & control
- ✗ Highest CapEx
- ✗ 5-year build; P&L impact Year 6
- ✗ Longest time-to-revenue
- ✗ High execution risk

Human Capital Strategy Decision

Building the workforce to support Plant Expansion

OUR DECISION: Full-Time Employees — Long-Term Investment in Our People

Option 1 — Contractors

✓ ADVANTAGES

- Lower upfront cost
- Quick onboarding process
- Flexible for ramp-up phase

✗ DISADVANTAGES

- No company culture alignment
- Reduced loyalty & commitment
- Limited long-term knowledge
- Higher turnover risk

Option 2 — Full-Time Employees ✓

✓ ADVANTAGES

- Long-term commitment & knowledge
- Aligned incentives with A&A goals
- Knowledge accumulates over time
- Supports Credo: security & dignity

✗ DISADVANTAGES

- Higher costs (benefits, training)
- Less flexibility to scale down

Financial Analysis

Key metrics from our chosen strategy — NTS · NPV · IRR · Cash Flow

\$22.3B

Net Present Value (NPV)

Discount Rate: 7%

Significant long-term value in today's \$

222%

Internal Rate of Return

vs. 7% hurdle rate

84.2%

Peak Market Share

Barbed suture market
56% starting position

~3 yrs

Payback Period

From initial investment

P&L SUMMARY (in \$M)

Metric	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Net Trade Sales (NTS)	\$6,188	\$6,916	\$7,744	\$8,693	\$9,727	\$10,921	\$12,269	\$13,826
Cost of Goods Sold	\$1,942: 31.38%	\$2,150: 31.09%	\$2,318: 29.93%	\$2,429: 27.94%	\$2,583: 26.55%	\$2,780: 25.46%	\$2,936: 23.93%	\$3,128: 22.62%
Gross Profit	\$4,247	\$4,766	\$5,425	\$6,264	\$7,144	\$8,142	\$9,332	\$10,698
Operating Expenses	\$2,611	\$3,014	\$3,067	\$3,259	\$3,471	\$3,678	\$3,908	\$4,077
CapEx / Investment	\$205	\$217	\$253	\$372	\$433	\$462	\$603	\$684
Free Cash Flow	\$1,074	\$1,209	\$1,811	\$2,140	\$2,804	\$3,355	\$3,908	\$4,989

Sensitivity Analysis

How changes in key input variables (pricing, growth, CapEx, market share, time-to-market) affect NPV, IRR, and Free Cash Flow across Base, Upside, and Downside scenarios

Base Case NPV: \$22.3B · IRR: 222% · Peak Market Share: 84.2% · Discount Rate: 7% · Payback: ~3 yrs · Break-Even: Yr 4

Variable	Base Case	Downside Scenario	Upside Scenario	Risk Level
Pricing (±10%)	Medium pricing; 80% reach; CM ~68%	Low pricing: −\$1.8B NPV	High pricing: +\$2.1B NPV	Medium
Market Growth Rate	7.5% CAGR; NPV \$22.3B; IRR 222%	+2% slower: −\$3.2B NPV	+2% faster: +\$3.8B NPV	High
CapEx Overrun	On budget; Mod. depreciation	+20% cost: −\$0.9B NPV	On schedule — base case	Medium
Market Share	56% start → 84.2% peak (10 yr)	Bear: 56% → 65% peak	Bull: 56% → 90% peak	High
Time to Market	Internal spiral: Yr 3 ramp; IRR 222%	+1 yr delay: −\$1.4B NPV	On schedule: base case	Medium

NPV & IRR Analysis

Discounted cash flow valuation and return analysis — validating financial viability at 7% hurdle rate

\$13.21B

Net Present Value (NPV)

Discount Rate: 7

222%

Internal Rate of Return (IRR)

Discount Rate Applied

32x

IRR / Hurdle Rate

IRR 222% vs. 7% required

\$19.1B

Cumul. Free Cash Flow (\$B)

Years 0-10 total

DISCOUNTED CASH FLOW & NPV BRIDGE (in \$M)

Metric	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Free Cash Flow (FCF)	\$1,257	\$1,347	\$1,212	\$1,415	\$1,434	\$1,674	\$1,775	\$1,870
Discounted Cash Flow	\$1,257	\$1,259	\$1,059	\$1,155	\$1,094	\$1,193	\$1,183	\$1,165
Cumul. Gross Profit	\$3,870	\$8,010	\$12,412	\$17,092	\$22,152	\$27,560	\$33,338	\$39,568
Cumul. Net Income	\$1,246	\$2,606	\$3,886	\$5,297	\$6,853	\$8,537	\$10,357	\$12,360
Total CapEx	\$212	\$402	\$581	\$793	\$1,057	\$1,326	\$1,581	\$1,881
NPV (Discounted)	\$1,257	\$2,516	\$3,575	\$4,730	\$5,824	\$7,017	\$8,200	\$9,365

Return on Investment (ROI)

Net profit efficiency of our total capital investment — $(\text{Net Income} - \text{CapEx}) / \text{CapEx}$

553

Overall ROI (%)

Discount Rate: \$19.7B

\$3.0B

Total CapEx Invested (\$B)

Cumul. Net Income (\$B)

\$16.7B

Net Profit After CapEx (\$B)

Net Income minus CapEx

488%

Year 0 ROI (Baseline)

Grows to 614% by Yr10

ANNUAL ROI BREAKDOWN (Net Income & CapEx in \$M)

Metric	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Net Income (\$M)	\$1,246	\$1,360	\$1,280	\$1,411	\$1,556	\$1,684	\$1,820	\$2,003
CapEx (\$M)	\$212	\$190	\$179	\$212	\$264	\$269	\$255	\$300
Annual ROI%	488%	614%	567%	488%	525%	614%	567%	614%
Net Profit (NI-CapEx)	\$1,034	\$1,170	\$1,101	\$1,199	\$1,291	\$1,415	\$1,565	\$1,703
Free Cash Flow	\$1,257	\$1,347	\$1,212	\$1,415	\$1,434	\$1,674	\$1,775	\$1,870
Cumul. Net Income	\$1,246	\$2,606	\$3,886	\$5,297	\$6,852	\$8,536	\$10,356	\$12,359

Contribution Margin Analysis

Per-unit profitability after variable costs — validating pricing strategy and gross efficiency

69.5

Avg. Contribution Margin (%)

Discount Rate: \$61.2B

\$88.1B

Cumul. Net Trade Sales (\$B)

Cumul. Gross Profit (\$B)

30.9%

Avg. COGS % of NTS

Low COGS = high margin

70.6%

Peak CM% (Year 10)

Improving over time

CONTRIBUTION MARGIN BY YEAR (in \$M)

Metric	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Net Trade Sales (\$M)	\$5,985	\$6,402	\$6,836	\$7,319	\$7,811	\$8,364	\$8,962	\$9,633
Gross Profit (\$M)	\$4,140	\$4,402	\$4,680	\$5,060	\$5,408	\$5,778	\$6,230	\$6,696
COGS (\$M)	\$1,845	\$2,000	\$2,157	\$2,259	\$2,403	\$2,586	\$2,731	\$2,937
Contrib. Margin %	69.2%	68.8%	68.5%	69.1%	69.2%	69.1%	69.5%	69.5%
Net Income (\$M)	\$1,360	\$1,280	\$1,411	\$1,556	\$1,684	\$1,820	\$2,003	\$2,207
CapEx (\$M)	\$190	\$179	\$212	\$264	\$269	\$255	\$300	\$309

Strategic Alignment & Why Our Strategy Wins

How every decision we made traces back to Johnson & Johnson's Credo

OUR DECISION: Option 4 — Develop Spiral Internally | Purchase IP of Symmetrical

CREDO ALIGNMENT

1 Patients First

Medium pricing keeps our barbed suture accessible to 80% of the target market — hospitals, surgical centers, and GPOs serving patients who need advanced wound closure solutions.

2 Our Employees

We will provide a sense of security, fulfillment and purpose. Hiring full-time employees — not contractors — ensures long-term job security, benefits, and dignity for our workforce. Knowledge builds over time; loyalty is mutual.

3 Our Community

Expanding the Cornelia, GA plant invests in the local community. Jobs, tax revenue, and infrastructure — not outsourcing to a third party or abandoning the facility.

4 Stockholders

When we operate according to these principles, stockholders should realize a fair return. NPV of \$[X] at [X]% discount rate. IRR of [X]% — [X]x the hurdle rate. ROI of [X]%. Payback in ~[X] years. Stockholders are rewarded because we did the first three right.

WHY OUR STRATEGY WINS

Internally Consistent

Every decision reinforces the others — Option 4 R&D, medium pricing, Silver/Gold marketing, plant expansion, and full-time employees all point to the same direction.

Financially Dominant

NPV: \$[X]M at [X]%. IRR: [X]% (hurdle rate: [X]%). Payback: ~[X] years. Revenue stream starts Year [X]. Positive across every scenario in our sensitivity analysis.

First-Mover in Spiral

No spiral suture exists on the market today. We own that IP, we own that clinical narrative, and we have a head start any competitor can require ~4–5 year horizon to close.

Stress Tested & Defensible

We modeled bear, base, and bull cases. In the worst downside scenario, all our financial metrics stay positive. This strategy is resilient — not just optimistic.

Thank You

Aegis & Aegis | Wound Closure Expansion Strategy | Rutgers UCC 2026